

Assignment 3

Deadline: 30 August 2026 11:30 PM

Submission link: <https://forms.gle/6NsNaBMpeYsDZqK38>

Solve any three problems:

Projection:

1. The projected point P' (44, 58, -195) for a point P (30, 50, -245) is given. The Projection Plane (PP) (XY plane parallel) is 200 units away from COP along Z axis. Find the COP.
2. For COP(3, 6, 10) find the projected point P' for a point P (-245, 50, 30) given projection plane PP (YZ parallel) is 180 units away from COP in X axis.
3. For COP(3, 5, 10) find the projected point P' for a point P (30, 245, 50) given projection plane PP (ZX parallel) is 190 units away from COP in Y axis.
4. For a point P (10, 20, -40), calculate the projected point P' using Cabinet Projection, where the orientation angle is 30° where the projection plane is at $x = 7$.
5. For COP (4, 5, 6) find the projected point P' for a point P (30, 245, 50). Given the projection plane PP $x+3y = -5$
6. Find the orthogonal projection of a point P (30, 245, 50) on the plane $x-y = 1$
7. For a point P (10, 20, -40), calculate the projected point P' using Orthographic Projection.
 - a. For the projection plane of XY
 - b. For the projection plane of $y = 13$
8. Why is Cavalier projection called full length projection?
9. Why is Cabinet projection called half length projection?

Color:

1. Why do we need Subtractive color model? (Hint: for printing, painting, producing active black color in printer)
2. CMY model (subtractive model) can handle printing task, then why do we need CMYK model?
3. Can we produce active black color in RGB model?
4. **Understanding 10-bit Color Depth:**

In an **8-bit** system, each color channel (Red, Green, and Blue) has $2^8 = 256$ possible levels (ranging from 0 to 255).

In a **10-bit** system, each color channel (Red, Green, and Blue) has $2^{10} = 1,024$ possible levels.

- The range for each color is **0 to 1,023**.

A photographer is editing a high-dynamic-range (HDR) photo of a sunset. The image is captured in **10-bit RGB format**. A specific pixel in the orange glow of the sun has the following raw 10-bit values:

- **Red:** 922
- **Green:** 461
- **Blue:** 102

Your Task:

1. Determine the 'normalized' RGB values (on a scale of 0 to 1).
 2. Convert this 10-bit RGB pixel into the **HSV** color space.
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5. Difference between color gamut and chromaticity diagram.